

StudyForge Sample Study Notes

Topic: Photosynthesis and Plant Biology

Photosynthesis is the process by which green plants, algae, and some bacteria convert light energy into chemical energy. In plants, it mainly occurs in the chloroplasts.

Key Points

- Chlorophyll absorbs light energy, especially in the blue and red regions of the spectrum.
- The light-dependent reactions take place in the thylakoid membranes of chloroplasts.
- The Calvin cycle takes place in the stroma and uses carbon dioxide to build carbohydrates.
- Water provides electrons and hydrogen ions, while oxygen is released as a by-product.
- Glucose stores chemical energy that can later be used by the plant.

Important Terms

Chloroplast: The organelle where photosynthesis occurs.

Chlorophyll: A pigment that absorbs light energy.

ATP: An energy-carrying molecule produced during the light reactions.

NADPH: An electron carrier used in the Calvin cycle.

Stomata: Small openings in leaves that allow gas exchange.

Practice Questions

1. Where does photosynthesis occur in a plant cell?
2. What role does chlorophyll play?
3. What gas is released during photosynthesis?
4. What is the purpose of the Calvin cycle?
5. Why is sunlight important to photosynthesis?